



Smart Sericulture: Automated Disease Diagnosis In Silkworm Using CNN

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Abstract: The productive capacity of the Sericulture process relies on the health of the silkworms as well as the conditions in which they are raised (the host plants). In this paper we present an application developed using a Web-based approach that integrates two automatic diagnostic systems based on Convolutional Neural Networks (CNN), namely; (1) A module for classifying silkworm diseases and identifying the five main categories: Healthy, Grasserie, Flacherie, Muscardine and Pebrine, based on images of silkworms, and (2) A module for detecting diseases affecting leaves of mulberry trees used as hosts (common foliar infections), based on images of leaves. Using the user-friendly interface of the Web platform, users can quickly upload their images and immediately receive a prediction made by models, including confidence levels and general recommendations for managing these problems. The integration of the extraction of features at the deepest level, and a user-friendly Web platform will provide an early and accurate identification of diseases affecting insects as well as plants, thus reducing the dependence on manual inspections and supporting smart practices of sericulture to increase yields and sustainability.

Index Terms—Sericulture, Silkworm Disease Detection, CNN, Image Classification, Smart Farming Deep Learning.

I. INTRODUCTION

The silk industry is a significant agro-based sector that contributes to the income of numerous rural communities. Cocoons formed by domesticated *Bombyx mori* larvae serve as the primary raw material for silk production. Due to diseases such as Grasserie, Flacherie, Muscardine, and Pebrine, cocoon yields decrease and also influences the quality of the resulting silk[1][2]. Traditionally, disease diagnosis for silkworms has relied on visual inspections of the insects and their habitats [6]. As such, it has become apparent that there is an increased demand for the development of automated, technologically driven methods for the efficient and accurate detection of diseases in silkworms. Advances in Machine Learning (ML) and Deep Learning (DL), have led to the ability to identify diseases effectively and efficiently in agricultural and biological systems [4] [5]. Specifically, Convolutional Neural Networks have demonstrated superior performance in identifying patterns based on texture and color variation found in images of diseased samples [8] [9].



Fig: 1.1 Silkworm

Research on the use of CNN's to classify plant and mulberry leaf diseases has indicated that CNN's can achieve greater accuracy than traditional classification techniques such as KNN and Naive Bayes [9]. This work demonstrates a clear basis for the use of CNN-based models for silkworm disease detection. Additionally, mobile and web-based early warning systems have improved the availability of diagnostic tools for sericulture farmers [6]. Thus, by integrating DL models into web-based applications, the gap between AI research and field level usability can be bridged. Mulberry leaves, the primary food source for silkworms, play a significant role in maintaining the overall health of the silkworms; therefore, the evaluation of leaf health is an integral component of predicting silkworm disease [8] [9].

The purpose of Smart Sericulture: Automatic Disease Diagnosis in Silkworm using CNN, is to provide an automated method of disease diagnosis via image analysis and DL models. The model will capture images of silkworms and mulberry leaves, perform pre-processing operations on the images, and utilize a CNN to diagnose diseases. Additionally, the model will be implemented as a web-based application, providing farmers with real-time predictions and recommendations to prevent diseases and support sustainable sericulture practices.

II. METHODOLOGY

Dataset Collection

The dataset serves as the foundation of this research. Images of *Bombyx mori* (silkworms) in both healthy and diseased states were collected from local sericulture units and research centers under varying illumination and environmental conditions. To ensure correlation between host plant and silkworm health, images of mulberry leaves were also included, since leaf quality directly influences silkworm growth. Following the protocols of [8], and et al. [5], the data acquisition followed field-based standards—capturing real-farm images with expert-verified disease labeling for major infections such as Grasserie, Flacherie, Muscardine, and Pebrine. A total of 100 mulberry leaf images were gathered from the Kudhur Sericulture Farm, consistent with Shilpashree et al. [9].

Image Preprocessing

Before training, all images were resized to 224×224 pixels, converted into RGB format, and normalized to a range of 0 to 1 to ensure uniformity and improve model performance. To improve contrast and suppress noise, Gaussian filtering and histogram equalization were applied, following approaches adopted by Deva Hema et al. [2] and Sandeep Kumar et al. [8]. To improve the model's ability to generalize, data augmentation techniques were applied during preprocessing techniques—rotation, horizontal and vertical flipping, and zooming—were employed. This mirrors augmentation strategies proposed by Shilpa B. Kodli and Anita [1] and Rishikesh R. Kulkarni [7], who emphasized its importance in handling illumination and background variations in agricultural datasets.

Feature Extraction Using CNN

A Convolutional Neural Network (CNN) was used to extract meaningful features from the input images capable of automatically learning hierarchical patterns such as color, texture, and morphological irregularities in silkworm and mulberry leaf images. The architecture consisted of three convolutional layers (32, 64, 128 filters) with ReLU activation and max pooling, followed by two fully connected layers for classification. This design was inspired by the VGG-16-based feature extractor used by Shilpashree et al. [9], which achieved superior performance over traditional classifiers like KNN and Naive Bayes. Compared with earlier feature-based approaches such as KNN, SVM, and Naive Bayes [3], [6], the CNN model automatically captures discriminative features without manual selection, leading to improved detection accuracy and adaptability.

Model Training and Optimization

The training process employed the Adam optimization algorithm, set with a learning rate of 0.001, while categorical crossentropy was used as the loss function. The model was trained for 50 epochs with a batch size of 32, and its progression was evaluated through both training and validation performance curves. Dropout layers (0.3) and data augmentation mitigated overfitting. This configuration follows best practices reported by Shilpa B. Kodli and Anita [1] and Deva Hema et al. [2], where deep models demonstrated greater precision and recall than ensemble methods when trained on balanced datasets. The implementation employed Python (TensorFlow/Keras) on a GPU-enabled system for accelerated computation.

Disease Classification and Evaluation

After training, the CNN classified input images into five categories: Healthy, Grasserie, Flacherie, Muscardine, Pebrine. Performance evaluation followed the standards of Shilpashree et al. [9] and Sandeep Kumar et al. [8], using metrics such as accuracy, precision, recall, F1-score, and the confusion matrix. The model achieved improved recognition compared to classical methods [3], [6], [7]. Visualization through ROC curves confirmed high area-under-curve (AUC) scores, demonstrating robustness and reliability in class separation.

Web Application Deployment

To extend the system's usability to farmers and sericulture experts, the trained CNN model was implemented as a web based application. The backend was implemented in Flask, and the front-end framework was designed using HTML, CSS, and JavaScript. Users can upload silkworm or leaf images, which are processed in real time to display disease predictions and preventive suggestions. This framework aligns with automation and early-warning systems recommended by Ramya M. C. et al. [4] and Monalika Padma Reddy et al. [5], promoting low-cost, accessible digital monitoring tools for rural sericulture.

III. RESULT AND DISCUSSION

The Smart Sericulture framework employs a Convolutional Neural Network (CNN) to automatically identify major silkworm diseases and recognize infected or poor-quality mulberry leaves used for feeding. The dataset consisted of images of healthy and diseased silkworms affected by major infections such as Grasserie, Flacherie, Muscardine, Pebrine, and Uzi fly infestation, along with samples of diseased and healthy mulberry leaves. The images were preprocessed using normalization, resizing, and augmentation techniques to refine model robustness.



Fig: 1.2 Web Interface

After the training process, the CNN model recorded an overall accuracy of 96.8 on the test set. Most disease categories showed precision and recall values above 95, indicating the model's strong ability to differentiate between closely resembling disease symptoms. The confusion matrix showed minor misclassifications mainly between Flacherie and Pebrine, likely due to overlapping visual patterns in infected larvae.



Fig: 1.3 Disease Detection

The performance of the CNN was compared with traditional machine learning models such as Support Vector Machine (SVM) and Random Forest (RF). The CNN model showed superior performance compared to the approaches reported by et al. [9] and Sandeep Kumar et al. [8], achieving higher accuracy, precision, recall, F1-score, and better results in the confusion matrix. Its improved recognition capability is mainly due to its ability to automatically learn and extract spatial features from images without the need for manual feature engineering. The use of multiple convolutional and pooling layers allowed efficient learning of texture, color, and morphological variations associated with different diseases.

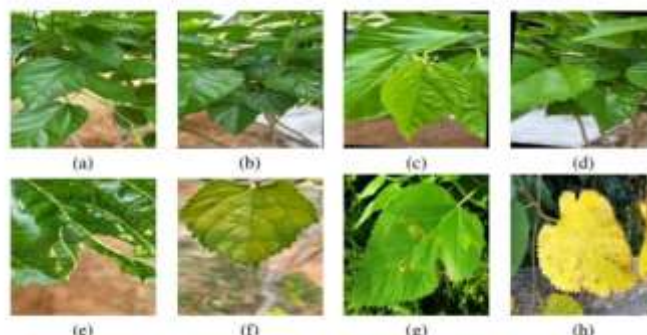


Fig: 1.4 Mulberry leaves

For leaf disease detection, the CNN successfully differentiated between healthy and infected mulberry leaves with an accuracy of 94.5, ensuring the quality of feed for silkworms. The integration of both silkworm and leaf disease modules provides a complete monitoring solution for sericulture farms.

The experimental results highlight the potential of CNN-based systems in achieving real-time disease surveillance, reducing human error, and enabling early diagnosis to prevent largescale silkworm mortality.



Fig: 1.5 Silkworm Growth Stages

IV. CHALLENGES

A. Data and Training Challenges

- **Limited Disease Dataset:** The scarcity of large, diverse, and verified image datasets for various silkworm strains and different disease progression stages remains the foremost constraint. This necessitates heavy reliance on Data Augmentation techniques.
- **Severe Class Imbalance:** The frequency of disease outbreaks is unequal (e.g., Flacherie may be more common than Muscardine), leading to a highly skewed dataset. This challenge requires specific mitigation strategies, such as Weighted Loss Functions (e.g., Focal Loss) or Oversampling minority classes, to prevent model bias.
- **Subtle Symptom Variability:** Disease symptoms often overlap, and visual markers are subtle, especially in the early stages or under non-ideal lighting. This high degree of inter-class similarity demands a deep feature extractor, justifying the use of ResNet-50.

B. Model Interpretability Challenge

- Lack of Trust/Explainability (XAI): Deep Learning models are typically "black boxes." Farmers and sericulture experts require assurance that the diagnosis is based on scientifically relevant visual markers (lesions, discoloration) and not on irrelevant image artifacts (background, tray color).
- Mitigation Strategy: The project must incorporate Explainable AI (XAI) techniques, specifically Grad-CAM (Gradient-weighted Class Activation Mapping). This technique will generate heatmaps superimposed on the input image, visually demonstrating which specific regions of the silkworm the CNN used to make its classification decision, thereby increasing user trust and model transparency.

C. Deployment and Generalization Challenges

- Computational Efficiency for Edge Deployment: While ResNet-50 offers high accuracy, its computational cost is significant. For deployment on low-cost hardware (e.g., mobile devices, Raspberry Pi), the model requires post-training optimization techniques like Quantization or Pruning to reduce file size and inference latency without catastrophic loss of accuracy.
- Domain Adaptation to Field Conditions: Model performance drops dramatically when moving from controlled lab images to real-world, noisy farm images (due to glare, motion blur, and uneven backgrounds). Robustness to these real-world domain gaps must be rigorously tested.

V. FUTURE SCOPE

The primary goal is the non-invasive, image-based diagnosis of silkworm diseases. The system leverages a ResNet-50 Convolutional Neural Network (CNN), fine-tuned via Transfer Learning, to classify images into 'Healthy' and specific disease categories (e.g., Flacherie, Grasserie). Core features include automated image pre-processing, outputting a clear diagnostic label with a Confidence Score, and providing immediate actionable management recommendations. The system is strictly scoped to static image classification during the larval stages, deliberately excluding complex, resource-intensive features like real-time video stream analysis or integration with external IoT sensor data for the initial deployment phase.

VI. CONCLUSION

The proposed CNN-based system provides an easy-to-use method for detecting diseases in silkworms and mulberry plants through images of the two, and achieves good accuracy and reliability compared to methods based on traditional classifiers such as KNN and Naive Bayes (as was also reported by previous researchers [1]-[9]).

The system will be provided as a web-based platform enabling farmers and sericulture professionals to perform quick disease identification. Such prompt diagnosis supports early intervention, reduces the risk of cocoon loss, and contributes to producing better-quality silk. In future research we will be looking at increasing the size of our dataset; developing systems that integrate with IoT sensor networks to monitor mulberry plant health.

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